

Dropbox acquires Endorse

Endorse is an app developer known for offering cashback deals to brand loyalists. Endorse was shut down recently, so Dropbox's buyout of it is a bit of a mystery

Rovio Accounts launched

Rovio Accounts is available for Android and iOS users, enabling users to save progress and sync across devices

Z87 Motherboards Test

HOW WE TESTED

Motherboard performance is completely a function of its components. The higher specced the components, the better performance you will get. For the same chipset, the variation in scores is not much as will be evident from our scores. The difference is within the margin of error. This can be easily understood by the fact that benchmark scores from a ₹10,000 board are very close to the benchmark scores from a ₹30,000 motherboard, give or take 5 percent. It ultimately boils down to the feature set you are getting with the motherboard, the build quality, ease of overclocking and the various ways in which you can overclock, among other factors. Keeping this in mind, we decided to not give any weightage to the performance scores.

Test Rig

Processor: Intel Core i7-4770K

SSD: Kingston HyperX 120GB SSD

RAM: 4x4GB Kingston HyperX @ 1866 MHz

Cooler: Cooler Master Hyper T4

Power Supply: Cooler Master Silent Pro 850W

OS: Windows 7 Ultimate x64

We ran the benchmarks and logged the scores for pure reference purposes. As you will see there is not much of a difference as the chipset is the same.

The layout of the board was scored, we checked if there was enough spacing around the CPU region to comfortably attach the CPU cooler and which way the SATA ports were pointing (those pointing up got a lesser score). We also noted the spacing between PCIe and PCI slots. If a dual slot card blocked the next PCIe slot, marks were deducted.

For build quality we noted the sturdiness of the heat sinks around the VRM regions and southbridge. Motherboards lacking a heatsink around the VRM region got lesser score, whereas those having an active cooling mechanism on the Southbridge scored higher. Board with all solid state capacitors got a higher score. Back panel I/O ports were noted and also the option to expand USB ports. Fan connectors, SATA connections and USB headers were also noted.

Overclocking is a key feature with Z87 boards. We overclocked most of the boards to at least 4.5GHz at the very least. Overclocking prowess can again vary and is subject to parameters such as what cooler you are using, how good the air-flow of your cabinet is, whether you are using air cooling or liquid cooling. We have played with just the basic settings such as BCLK and multipliers, but you can also do voltage tweaking to get a higher overclock. Features such as overclocking utilities, on-board chip additions, were taken into consideration. A board having more options to overclock and tweak got a higher score.

Also the BIOS user experience was noted. Since most BIOSes these days have a graphical interface, we laid stress on the ease of navigation with the help of home screens, profile settings, shortcut keys and ease of changing settings.

We did a thorough checking and testing (wherever applicable) of the feature set offered by a motherboard and based on the price have decided on the winners.

the VRMs as well as the southbridge are quite impressive. It comes with two PCIe x16 slots which support only CrossFireX with x8/x8 speeds. The two upwards facing SATA ports are a bit of an eyesore on this otherwise well laid out board.

MSI Z87-G43 offers a utility called Command Centre which can be used on the desktop as well as an app on your Android smartphones. It allows you to monitor your system settings and tweak them from either platforms. Although the MSI board does not have a OC Genie physical button, you do get a virtual OC Genie button in the BIOS which overclocks your system. The best way to overclock is still via the ClickBIOS 4 screen. The A-Tuning utility of the ASRock board isn't as elaborate as the Command Centre, but gets the work done. Overclocking using

the OC Tweaker is a breeze – you can either do it using the presets or you can manually tweak settings with the sliders. Overclocking via the BIOS is also pretty quick. The Biostar utility T-Overclocker wasn't as impressive as the ones seen on other boards in this segment. Just like MSI, the Biostar board also comes with a BIO-Remote 2 app which allows you to not just overclock from an app, but also use your mobile device as a remote control to work with your system.

We could overclock the processor on each of the boards till 4.6GHz getting a comfortable 20 percent boost in Cinebench R11.5 readings. the scores were nearly similar except in gaming where the ASRock Z87 Pro3 gave comparatively lower scores – 5 fps lower.

10k-15k

This category will be important, as we feel most volumes will be selling within this price bracket for the new platform.

Let us start off with the Intel DZ87KLT-75K which is the last motherboard manufactured by Intel. This series 8 board is Intel's swansong as far as desktop motherboards go. As compared to the Z77 chipset based Intel DZ77GA-70K, the DZ87KLT-75K looks



ASRock Z87 Extreme 4

quite different. For starters, the skull embossed heatsink on the southbridge and blue-coloured PCIe slots are absent. Heatsinks around the VRM section are better than the last generation. While there are eight SATA ports, dedicated power/reset buttons and dual two-digit POST code readouts, we were not impressed with the display outs – just one HDMI port. It is the only board in this segment which comes with a Thunderbolt port. Along with the on-board connections, it also bundles in a WiFi / Bluetooth module. The board is well laid out with enough clearance around the CPU region and enough spacing between successive PCIe x16 slots.

Coming to the two ASRock boards – Z87 Extreme 4 and Z87 Extreme 6. Both the brown coloured boards